

COMPLETE SYSTEM GUIDE



Architecture • Protocols • APIs • Dashboard • Blockchain

VERSION 2.0 • 2026

API ENDPOINT: <https://api.alshumookh-pay.com>

BLOCKCHAIN: Ethereum Mainnet • Base • Tron TRC-20

SIG TOKEN: 0xc2AC880E474c3576CC3afB7c560E402ce24d5B37

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SYSTEM OVERVIEW

Company Identity · Tech Stack · Key Metrics

1.1 Company Identity

ALSHUMOOKH GLOBAL BANKING FINANCE & CREDIT is an international financial technology institution operating a full-stack digital settlement platform. The system bridges fiat banking, stablecoin settlement, and proprietary token issuance on Ethereum Mainnet. All transactions are cryptographically audited and reconciled in real-time via blockchain confirmation monitoring.

Field	Value
Company	Al Shumookh International Group
Platform	ALSHUMOOKH GLOBAL BANKING FINANCE & CREDIT
API Domain	https://api.alshumookh-pay.com
API Version	v1 (prefix: /api/v1)
Environment	Production (Ethereum Mainnet · Chain ID: 1)
Notification Email	info@alshumookhgroup.com
System Email	no-reply@alshumookhgroup.com
Framework	FastAPI (Python 3.12) — Async/Await Architecture
Database	PostgreSQL (async via SQLAlchemy + aiomysql for dev)
Task Queue	Celery + Redis
Deployment	Render.com — auto-deploy from GitHub main branch

1.2 Technology Stack

Layer	Technology	Purpose
Web Framework	FastAPI 0.100+	Async REST API, OpenAPI docs
ORM	SQLAlchemy 2.0 (async)	Database models & queries
Task Queue	Celery + Redis	Background jobs, monitoring tasks
Blockchain	web3.py	Ethereum/Base contract interaction
Signing	eth-account	Transaction signing
Webhook Verification	HMAC-SHA256	Alchemy, Stripe, MoonPay
Auth	OAuth 2.0 + API Keys	Enterprise & partner access
PDF Generation	reportlab	System reports & documents

Encryption	JWE (RSA-OAEP)	Payload encryption
Signature	JWS (RS256)	Payload signing
mTLS	Client Certificates	High-security counterparty auth
Email	SMTP/TLS	Notifications
Monitoring	Sentry DSN	Error tracking

1.3 Supported Assets & Networks

The platform natively supports four crypto assets across three blockchain networks. All settlement wallets are hardware-secured Ledger devices. The operator (hot) wallet is MetaMask, used only for signing and broadcasting — never for custody.

Asset	Symbol	Network	Standard	Role
Tether USD	USDT	Ethereum / Tron	ERC-20 / TRC-20	Primary Settlement
USD Coin	USDC	Ethereum / Base	ERC-20	Inbound / Circle
Ether	ETH	Ethereum	Native	Gas Fees
AI Shumookh Token	SIG	Ethereum	ERC-20	Proprietary Token

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PAYMENT PROVIDERS

Coinbase · MoonPay · Circle · Stripe · Onramper

2.1 Coinbase Commerce / CDP

Coinbase Developer Platform (CDP) provides both on-ramp services and institutional settlement. The system uses the Coinbase Commerce API for order creation, Coinbase Advanced Trade for portfolio tracking, and CDP On-ramp widget for fiat-to-crypto conversion. Webhook events are verified via HMAC-SHA256 signature before processing.

Config Key	Value / Description
API Host	api.cdp.coinbase.com
Default Currency	USD (fiat)
Default Crypto	USDC
Default Network	Ethereum
On-ramp Base URL	https://api.cdp.coinbase.com
Redirect URL	https://api.alshumookh-pay.com/pay/success
Webhook Path	POST /api/v1/webhooks/coinbase
Settlement Path	POST /webhooks/coinbase

2.2 MoonPay

MoonPay handles fiat-to-crypto on-ramp for end customers. The platform creates MoonPay customers via API, generates widget URLs for the checkout flow, and processes webhook notifications when purchases complete. MoonPay supports card payments (Visa/MC), bank transfers, and local payment methods across 160+ countries.

Config Key	Value / Description
API Base URL	https://api.moonpay.com
Widget URL	https://buy.moonpay.com
Create Customer Path	/v1/deposit-customers/api-key
Webhook Path	POST /api/v1/webhooks/moonpay
Webhook Verification	HMAC-SHA256
Dashboard Page	/dashboard/payments#moonpay

2.3 Circle USDC

Circle provides programmable wallets for USDC settlement. The integration uses Circle's Web3 Services API with wallet sets for multi-wallet management. Circle USDC wallets receive institutional USDC payments and

reconcile them against pending settlement payloads.

Config Key	Value / Description
API Key	CIRCLE_API_KEY (env)
Entity Secret	CIRCLE_ENTITY_SECRET (env)
Wallet Set ID	CIRCLE_WALLET_SET_ID (env)
Circle Wallet ID	CIRCLE_WALLET_ID (env)
Circle Wallet Address	CIRCLE_WALLET_ADDRESS (env)
Webhook Path	POST /api/v1/webhooks/circle
Dashboard Page	/dashboard/payments#circle

2.4 Stripe

Stripe handles traditional card payment processing. The system creates Stripe Checkout sessions for fiat payments, receives webhook notifications via signature-verified events, and maps completed charges to pending payment orders. Stripe is also used for invoice generation and subscription management.

Config Key	Value / Description
API Base URL	https://api.stripe.com/v1
Webhook Path	POST /api/v1/webhooks/stripe
Success URL	STRIPE_SUCCESS_URL (env)
Cancel URL	STRIPE_CANCEL_URL (env)
Webhook Verification	Stripe-Signature header
Dashboard Page	/dashboard/stripe

2.5 Onramper (Multi-Provider On-Ramp)

Onramper aggregates multiple fiat on-ramp providers behind a single widget. The system embeds the Onramper widget for customers who need payment method flexibility. Default crypto is USDC, default fiat is USD.

Config Key	Value
Widget Base URL	https://buy.onramper.com
Default Crypto	USDC
Default Fiat	USD

3

BLOCKCHAIN NETWORKS

Ethereum · Base · Tron · Wallets · Contracts

3.1 Network Overview

Network	Chain ID	Standard	Token Support	RPC Provider	Role
Ethereum Mainnet	1	ERC-20	USDT, USDC, ETH, SIG	Alchemy	Primary
Base Mainnet	8453	ERC-20	USDC	Alchemy	Secondary
Tron Mainnet	N/A	TRC-20	USDT	TronGrid	Tertiary

3.2 Smart Contract Addresses

Contract / Token	Network	Address
SIG Token (AI Shumookh)	Ethereum	0xc2AC880E474c3576CC3afB7c560E402ce24d5B37
USDT (Tether USD)	Ethereum	0xdAC17F958D2ee523a2206206994597C13D831ec7
USDT (TRC-20)	Tron	TR7NHqjeKQxGTCi8q8ZY4pL8otSzgjLj6t
M1 Fund Token	Ethereum	0xd999DB972BBDC2C13a9595A1474A04F5e59169a5

3.3 Wallet Architecture

The system uses a two-wallet architecture separating custody from operations. The Treasury Wallet (Ledger hardware) receives all inbound funds and holds reserves — it is NEVER exposed to the API. The Operator Wallet (MetaMask software) only signs and broadcasts transactions. This design ensures that even a complete API compromise cannot drain the treasury.

Wallet Role	Type	Address	Function
Treasury / Ledger	Ledger Hardware	0xBD682cfD8382a90adfDd6745780D3D7959c4d939	RECEIVES funds, custody
Operator	MetaMask Software	0x620a850efe2c97A02560d9bce9978639Ed232BE2	SIGNS & broadcasts txs
Tron Treasury	Hardware	Configured via env	TRC-20 USDT receipt
Circle Wallet	Circle Programmable	Configured via env	USDC settlement receipt

3.4 Alchemy Integration

Alchemy is the primary blockchain RPC and webhook provider. It monitors the treasury wallet for all inbound transfers (USDT, USDC, SIG) and fires real-time webhook notifications to the platform. The system verifies

every Alchemy webhook using HMAC-SHA256 with the `ALCHEMY_WEBHOOK_SIGNING_KEY`. Alchemy also provides ETH and Base RPC endpoints for settlement transaction verification.

Parameter	Value
Network	eth-mainnet
Webhook Path (events)	POST /webhooks/alchemy
Webhook Path (settlement)	POST /api/v1/webhooks/alchemy
Verification	HMAC-SHA256 (X-Alchemy-Signature header)
Confirmations Required	12 blocks
Monitoring Interval	60 seconds
Etherscan Explorer	https://etherscan.io/tx/{tx_hash}

4

INBOUND SETTLEMENT PIPELINE

Payload Ingestion · Verification · Reconciliation

4.1 Pipeline Overview

The inbound settlement pipeline receives crypto payment notifications from counterparties, verifies them on-chain via Alchemy, reconciles them against pending ExternalPayload records, and triggers downstream processing (tokenization, outbound transfer, or manual review). The pipeline is fully audited at every step.

Step	Action	Status Transition
1	Counterparty POSTs payload to /api/v1/payloads/ingest	RECEIVED
2	System parses payload: amount, asset, tx_hash, network	PARSED
3	Awaiting blockchain transaction hash from counterparty	AWAITING_TX_HASH
4	Alchemy webhook fires when transfer detected	ALCHEMY_PENDING
5	Alchemy confirms the transfer (amount, asset, wallet)	ALCHEMY_VERIFIED
6	System waits for 12 block confirmations on Ethereum	ON_CHAIN_CONFIRMED
7	Payload matched to pending settlement — funds reconciled	RECONCILED
8	Failure at any step — admin review required	FAILED / MANUAL_REVIEW

4.2 ExternalPayload Model Fields

Every inbound payload is stored as an ExternalPayload record with full audit trail. Key fields include the parsed transaction reference, blockchain details, security verification flags, and review decisions.

Field	Type	Description
id	UUID	Unique payload identifier
api_client_id	UUID	Which counterparty submitted this payload
raw_payload	Text	Original JSON body, unmodified
tx_hash	String(128)	Blockchain transaction hash
sender_wallet	String(128)	Source wallet address
receiver_wallet	String(128)	Destination wallet (our treasury)
amount	Decimal(30,18)	Transfer amount in token units
asset	String(32)	Token symbol: USDT, USDC, SIG
network_name	String(32)	ethereum, base, tron
verification_status	Enum	RECEIVED→PARSED→ALCHEMY_VERIFIED→RECONCILED
security_level	String	hmac_verified / api_key_only / unsigned

jws_verified	Boolean	JWS signature verified (enterprise tier)
jwe_decrypted	Boolean	JWE payload decrypted (enterprise tier)
mtls_verified	Boolean	mTLS client cert verified (enterprise tier)
review_decision	String	APPROVED / REJECTED / HOLD
review_note	Text	Admin review comments
block_number	BigInt	Block number of the confirmed transaction
confirmations	Integer	Number of blockchain confirmations received

4.3 FNFCU V1.0 — Cash Transfer Protocol

FNFCU V1.0 (Financial Network Fund Clearance Unit) is the proprietary inbound cash transfer protocol used by institutional counterparties for structured settlement payloads. It uses a JSON schema with mandatory fields: transaction reference, amount, asset type, network, sender information, and authorization code. HMAC signature is required for enterprise counterparties. JWS signing and JWE encryption are available for the highest security tier.

Security Tier	Authentication Method	Required Fields
Basic	API Key (X-API-Key header)	tx_ref, amount, asset, network
Standard	API Key + HMAC-SHA256	Above + X-HMAC-Signature
Enterprise	OAuth 2.0 + JWS	Above + signed JWT payload
Maximum	mTLS + JWE + JWS	Above + encrypted + cert verified

5

OUTBOUND TRANSFER ENGINE

USDT Transfers · Approval Workflow · Confirmation Tracking

5.1 Transfer Lifecycle

Every outbound USDT transfer goes through a mandatory approval workflow before broadcasting. Transfers are initiated by the system (auto-payout) or by admin action, then approved via the dashboard, broadcast to the blockchain, and tracked until confirmed.

Status	Description	Next Action
PENDING	Transfer queued, awaiting review	Admin reviews and approves
AWAITING_APPROVAL	Sent for explicit approval	Admin approves or cancels
APPROVED	Admin approved, ready to broadcast	System broadcasts
BROADCASTING	Transaction submitted to mempool	Await confirmation
PENDING_CONFIRMATION	Tx in mempool, awaiting blocks	Monitor block confirmations
CONFIRMED	Minimum confirmations received	Mark as completed
COMPLETED	Transfer fully settled	Trigger callback if set
FAILED	Broadcast or confirmation failed	Retry or manual intervention
CANCELLED	Cancelled by admin before broadcast	No further action

5.2 Transfer Record Fields

Field	Description
network	Destination blockchain: ethereum, base, tron
asset	Token: USDT (default), USDC, SIG
from_address	Operator wallet address (signing key)
to_address	Recipient wallet address
amount	Transfer amount (up to 6 decimal places)
tx_hash	Blockchain transaction hash once broadcast
gas_used	Gas consumed (Ethereum/Base only)
block_number	Block in which the tx was confirmed
confirmations	Current confirmation count
callback_url	Optional URL to notify on completion
retry_count	Automatic retry counter
explorer_url	Link to Etherscan/Tronscan transaction

6

M1 TOKENIZATION ENGINE

EUR → USDT → SIG Conversion Pipeline

6.1 Tokenization Concept

The M1 Tokenization Engine converts EUR wire transfers (from the M1 Fund) into on-chain USDT or SIG tokens. The pipeline fetches real-time EUR/USD exchange rates, calculates the USDT equivalent, issues the corresponding SIG tokens at the fixed M1 issuance price, and dispatches the tokens to the destination wallet via an OutboundTransfer.

Tokenization Formula:

$$\text{EUR Amount} \times \text{EUR/USD Rate} \div \text{SIG_M1_PRICE_USD} = \text{SIG Tokens Issued}$$

Example: 23,085,000 EUR × 1.1537 USD/EUR ÷ 0.053266 USD/SIG = 500,000,000 SIG

Parameter	Value / Description
SIG M1 Issuance Price	\$0.053266 USD per SIG (fixed)
Tokenization Target Assets	USDT, SIG
Supported Input Currency	EUR (wire transfer), USDT (direct)
FX Provider	Configurable (ECB / Fixer / Manual)
Token Decimals	18 (ERC-20 standard)
Destination Network	Ethereum Mainnet
Job Status Flow	QUEUED → FX_FETCHED → CONVERTING → SENDING → COMPLETED

6.2 M1TokenizationJob Record

Each tokenization event is tracked as an M1TokenizationJob record with complete audit trail from EUR receipt through to on-chain confirmation.

Field	Description
sender_reference	SWIFT/wire reference from the sender
sender_name	Name of the EUR wire sender
sender_iban	Sender IBAN (for compliance)
eur_amount	EUR amount received
fx_rate_eur_usd	EUR/USD rate at time of conversion
usd_amount	Calculated USD equivalent
usdt_amount	USDT to be sent on-chain
network	Target blockchain (ethereum)

destination_wallet	Recipient wallet address
outbound_transfer_id	Linked OutboundTransfer ID
status	QUEUED / FX_FETCHED / CONVERTING / SENDING / COMPLETED

7

M1 RESERVE MODULE

Reserve Snapshots · Mint/Redeem · Oracle · Audit

7.1 Reserve Architecture

The M1 Reserve Module maintains an isolated ledger proving that all issued M1 Fund tokens (M1F) and SIG tokens are backed by real assets. Each reserve snapshot is cryptographically hashed and timestamped. Mint and redeem requests require multi-signature approval and are recorded on-chain.

M1FundReserve

Field	Description
fund_id	Unique fund identifier
total_reserve_value	Total USD value of backing assets
tokenized_value	USD value already tokenized
issued_tokens	Total tokens in circulation
available_to_mint	Remaining mintable tokens
backing_ratio	Collateralization ratio (e.g. 100%)
proof_document_hash	SHA-256 of proof document
valuation_date	Date of last independent valuation

M1ReserveSnapshot

Field	Description
fund_id	Fund reference
total_reserve_value	Reserve at snapshot time
issued_tokens	Tokens in circulation
backing_ratio	Collateralization ratio
proof_document_hash	Immutable proof hash
approved_by	Approving officer ID

7.2 Oracle & Public Verification

The M1 Oracle publishes real-time reserve data to any authorized reader. Every oracle read is logged in M1OracleRead with IP, client_id, and a cryptographic hash of the response. This provides a tamper-evident public audit trail that can be verified by third parties.

7.3 Blockchain Confirmations

M1BlockchainConfirmation records link every mint and redeem request to its on-chain transaction. The verification_status field tracks whether the transaction has been independently verified on-chain beyond just being recorded in the system database.

8

SIG TOKEN

AI Shumookh International Group Token · ERC-20 · Uniswap

8.1 Token Specifications

Parameter	Value
Full Name	AI Shumookh International Group Token
Symbol	SIG
Standard	ERC-20
Network	Ethereum Mainnet (Chain ID: 1)
Contract Address	0xc2AC880E474c3576CC3afB7c560E402ce24d5B37
Decimals	18
M1 Issuance Price	\$0.053266 USD per SIG
Total Supply (M1 allocation)	500,000,000 SIG (for 23,085,000 EUR ÷ 0.053266)
Burn Address	0x00
Explorer	https://etherscan.io/token/0xc2AC880E474c3576CC3afB7c560E402ce24d5B37

8.2 Uniswap V2 Liquidity Pool

SIG has a Uniswap V2 liquidity pool on Ethereum Mainnet paired with USDC. The pool is currently owned 100% by ALSHUMOOKH. Due to the pool's liquidity size, large swaps result in significant price impact. The recommended approach is to use the M1 Tokenization Engine for institutional-size SIG issuance, and reserve the Uniswap pool for smaller market trades.

Parameter	Value
Pool Pair	USDC / SIG
Protocol	Uniswap V2 (Ethereum Mainnet)
Pool Owner	ALSHUMOOKH (100%)
Slippage Tolerance	25% (recommended for current liquidity)
Warning	Very high slippage warning at current pool size
Migration Option	V2 → V3 via Uniswap Migrate button (concentrated liquidity)
V3 Benefit	Same liquidity, narrower price range, better capital efficiency
Buy Method	Uniswap.org — Sell: USDC / Buy: SIG
Swap Route	USDC → SIG (ETH→SIG route does NOT exist in V2)

8.3 Token Roles in the Platform

- › Settlement receipt: Counterparties can send SIG as settlement payment
- › M1 issuance: SIG is issued to investors converting EUR via M1 Tokenization Engine
- › Profit distribution: SIG distributed via the Profit Distributor dashboard page
- › Reserve backing: SIG holdings tracked in M1 Reserve module
- › Market liquidity: USDC/SIG Uniswap pool enables open-market trading

9

SECURITY ARCHITECTURE

WAF · Rate Limits · IP Controls · Challenge Mode

9.1 WAF Middleware

The Web Application Firewall (WAF) middleware is the first layer of protection. Every inbound request is scored based on User-Agent, IP reputation, country, and request patterns. Suspicious scores accumulate and trigger challenge mode at threshold 3+. Certain trusted paths are exempt from WAF checks.

Exempt Path	Reason
/api/v1/oauth/token	OAuth token endpoint
/api/v1/payloads/ingest	Settlement payload ingestion
/api/v1/payloads/schema	Schema discovery
/api/v1/webhooks/stripe	Stripe webhook receiver
/api/v1/webhooks/alchemy	Alchemy webhook receiver
/api/v1/webhooks/circle	Circle webhook receiver
/api/v1/webhooks/coinbase	Coinbase webhook receiver
/api/v1/webhooks/moonpay	MoonPay webhook receiver
/webhooks/alchemy	Alchemy events webhook
/webhooks/coinbase	Coinbase events webhook
/webhooks/moonpay	MoonPay events webhook
/api/v1/transfer-request	Transfer request endpoint
/api/v1/transfers	Outbound transfers endpoint

9.2 Risk Scoring System

Event	Risk Score Added	Trigger
Suspicious User-Agent	+2	Python, curl, wget, go-http detected
Missing User-Agent	+1	No UA header
Suspicious Country	+1	Country in blocked list
Probe attempt	+2	Non-existent endpoint accessed
Challenge mode active	Block	Total score $\geq$ 3
IP ban	Block all	Probe threshold exceeded (5 probes/10min)
Ban duration	—	900 seconds (15 minutes)

9.3 Protected API Prefixes

The audit middleware logs every request to these protected paths with full context: client identity, IP, method, endpoint, status code, transaction ID, and request ID.

- › /api/v1/admin
- › /api/v1/oauth
- › /api/v1/payloads
- › /api/v1/transfer-request
- › /api/v1/transfers
- › /api/v1/transfer/
- › /api/v1/transactions
- › /api/v1/payments/ledger/status

9.4 Rate Limiting

Limit	Value
Global rate limit window	60 seconds
Global max requests per window	240 requests
Security probe window	600 seconds (10 minutes)
Probe threshold before ban	5 probes
Ban duration	900 seconds (15 minutes)
Silent probe blocks	Enabled (probes return no indication)

10

WEBHOOKS & EVENTS

Real-Time Event Processing · Signature Verification

10.1 Webhook Endpoints

Provider	Path	Verification Method	Function
Alchemy (Events)	POST /webhooks/alchemy	HMAC-SHA256	Monitor treasury wallet transfers
Alchemy (Settlement)	POST /api/v1/webhooks/alchemy	HMAC-SHA256	Settlement verification
Stripe	POST /api/v1/webhooks/stripe	Stripe-Signature	Payment order updates
Coinbase	POST /api/v1/webhooks/coinbase	HMAC-SHA256	Order completion events
Coinbase (Events)	POST /webhooks/coinbase	HMAC-SHA256	On-ramp events
MoonPay	POST /api/v1/webhooks/moonpay	HMAC-SHA256	Purchase completion
MoonPay (Events)	POST /webhooks/moonpay	HMAC-SHA256	On-ramp events
Circle	POST /api/v1/webhooks/circle	API verification	USDC wallet events

10.2 Alchemy Webhook Processing

When an Alchemy webhook fires, the system: (1) verifies the HMAC-SHA256 signature against the raw body using the `ALCHEMY_WEBHOOK_SIGNING_KEY`; (2) parses the activity array for token transfers to the treasury wallet; (3) calls `_match_external_payloads()` to find matching pending payloads; (4) updates the `ExternalPayload verification_status`; (5) logs the event to the `AuditLog`; (6) triggers downstream processing if matched.

10.3 WAF Exemption for Webhooks

All webhook paths are exempt from WAF security checks. This is critical because providers like Alchemy use automated HTTP clients (e.g., `python-requests`, `aiohttp`) that trigger the `SECURITY_USER_AGENT_FLAGGED` event. The exemption ensures webhook delivery is never blocked by the WAF while signature verification still provides cryptographic security.

11

OAUTH & API ACCESS

OAuth 2.0 · API Keys · HMAC · JWS / JWE / mTLS

11.1 Authentication Tiers

Tier	Method	Headers Required	Use Case
Basic	API Key	X-API-Key	Simple counterparty integration
Standard	API Key + HMAC	X-API-Key + X-HMAC-Signature + Idempotency-Key	Production payloads
OAuth	Bearer Token	Authorization: Bearer {token}	Enterprise OAuth clients
Enterprise	OAuth + JWS	Above + JWS-signed payload body	High-value settlement
Maximum	mTLS + JWE + JWS	Client certificate + encrypted payload	Institutional tier

11.2 OAuth 2.0 Configuration

Parameter	Value
Token Endpoint	POST /api/v1/oauth/token
Grant Type	client_credentials
Issuer	alshumookh-settlement-api
Audience	alshumookh-settlement
Token TTL	900 seconds (15 minutes)
Encryption	JWE (RSA-OAEP) with private key from SETTLEMENT_JWE_PRIVATE_KEY_PEM

11.3 ApiClient Model

Each counterparty is represented by an ApiClient record. Security flags can be toggled per client to require increasingly strict authentication.

Field	Description
name	Counterparty company name
api_key_hash	Bcrypt hash of the API key
hmac_secret	HMAC-SHA256 signing secret
hmac_required	Must include valid HMAC signature
oauth_required	Must use OAuth 2.0 Bearer token

jws_required	Payload must be JWS-signed
jwe_required	Payload must be JWE-encrypted
mtls_required	Client certificate required
allowed_ips	IP whitelist (JSON array)
egress_ip	Expected source IP for this client
bridge_contract_address	Optional smart contract bridge address

12

AUDIT & MONITORING

Audit Middleware · Event Types · Real-Time Dashboard

12.1 Audit Middleware

The audit middleware intercepts every request to protected endpoints and creates an AuditLog entry with: client identity, endpoint, HTTP method, client IP, user agent, response status code, transaction ID, request ID, and any error messages. The X-Request-ID header is set on all audited responses for traceability.

12.2 Audit Event Types

Event Type	Description
API_REQUEST	Every API call to protected endpoints
PAYLOAD_RECEIVED	New settlement payload ingested
PAYLOAD_VERIFIED	Alchemy on-chain verification completed
PAYLOAD_RECONCILED	Payload matched and funds reconciled
TRANSFER_INITIATED	Outbound USDT transfer created
TRANSFER_APPROVED	Admin approved outbound transfer
TRANSFER_BROADCAST	Transaction submitted to mempool
TRANSFER_CONFIRMED	12+ confirmations received
TOKENIZATION_STARTED	M1 EUR→USDT tokenization job started
TOKENIZATION_COMPLETED	Tokenization job completed
SECURITY_USER_AGENT_FLAGGED	Suspicious user agent detected
SECURITY_CHALLENGE_MODE	IP entered challenge mode
SECURITY_IP_BANNED	IP banned after probe threshold exceeded
WEBHOOK_RECEIVED	Webhook event received from provider
WEBHOOK_VERIFIED	Webhook signature verified successfully
OAUTH_TOKEN_ISSUED	OAuth 2.0 access token issued
ADMIN_ACTION	Admin performed a privileged action

12.3 Live Monitoring Page

The Live Monitoring dashboard (/dashboard/monitoring) provides real-time system health metrics including: pending payloads count, active outbound transfers, pending M1 tokenization jobs, recent security events, blockchain confirmation status, and provider webhook delivery rates.

13

DASHBOARD — ALL 21 PAGES

Complete Guide to Every Page and Every Button

13.1 Sidebar Navigation Overview

The dashboard sidebar contains 21 navigation links covering the full operational surface of the platform. Each page is role-protected and accessible via the admin API key. The sidebar is rendered server-side and available at /dashboard.

#	Path & Name	Purpose
1	■ Overview /dashboard	System health, recent events, quick stats, revenue summary
2	■ Orders /dashboard/orders	All payment orders across all providers — filter, search, view detail, export
3	■ Settlement Payloads /dashboard/payloads	Inbound ExternalPayload records — review, verify, approve, reject, hold
4	■ Outbound Transfers /dashboard/transfers	USDT outbound transfer queue — review, approve, monitor confirmations
5	■ M1 Tokenization /dashboard/tokenization	EUR→USDT→SIG job queue — view jobs, trigger conversion, monitor status
6	■ M1 Reserve /dashboard/m1-reserve	Reserve balance, backing ratio, snapshots, mint/redeem requests
7	■ Live Monitoring /dashboard/monitoring	Real-time system health — blockchain confirmations, webhook rates, alerts
8	■ Payments /dashboard/payments	Multi-provider payment hub — Coinbase, MoonPay, Circle, Direct Crypto
9	■ MoonPay /dashboard/payments#moonpay	MoonPay on-ramp orders, customer creation, webhook events
10	■ Circle USDC /dashboard/payments#circle	Circle wallet balance, USDC transfers, settlement matching
11	■ Direct Crypto /dashboard/payments#direct	Direct crypto payment links, QR codes, wallet addresses
12	■ Stripe /dashboard/stripe	Stripe checkout sessions, charges, webhooks, payout schedule
13	■ Alchemy Events /dashboard/alchemy	All Alchemy webhook events, transfer activity, wallet monitoring
14	■ Counterparties /dashboard/counterparties	API clients — create, configure, manage security tiers
15	■ Security /dashboard/security	WAF events, banned IPs, risk scores, challenge mode log

16	■ Documents /dashboard/documents	Transaction documents, uploaded evidence files, proof documents
17	■ Reports /dashboard/reports	Financial reports, reconciliation summaries, export to CSV/PDF
18	■ Audit Logs /dashboard/logs	Full audit trail — all events, filters by event type, date, client
19	■ Profit Distributor /dashboard/distributor	Distribute SIG profits to holders — set percentage, amount, timestamp
20	■ Top-Up Engine /dashboard/topup	Prepaid card management — wallets, cards, top-up transactions
21	■ SWIFT Terminal /swift	SWIFT message composer and payment instruction sender

13.2 Dashboard Buttons — Complete Reference

The following table lists every action button found across all 21 dashboard pages, the pages where each appears, and its exact function.

Button	Found On	Action
■ Approve	<i>Payloads, Transfers, M1 Reserve</i>	Approve the selected record and advance to next pipeline stage
■ Reject	<i>Payloads, Transfers</i>	Reject and mark the record as failed with optional rejection note
■ Hold	<i>Payloads</i>	Place record on manual hold pending further information
Save Note	<i>Payloads</i>	Save a review note/comment against the selected payload
Verify with Alchemy	<i>Payloads</i>	Manually trigger blockchain verification via Alchemy API
Manual Review	<i>Payloads</i>	Flag record for manual compliance review
Print Private Report ■	<i>Payloads, Reports</i>	Generate and print encrypted private report for this record
Clear	<i>Security</i>	Clear challenge mode or reset risk score for selected IP
Set %	<i>Profit Distributor</i>	Set the percentage of profits to distribute
Set Amount	<i>Profit Distributor</i>	Set a fixed SIG amount for distribution
Set Timestamp	<i>Profit Distributor</i>	Schedule distribution at a specific UTC timestamp
Export CSV	<i>Reports, Logs</i>	Export filtered records to CSV file
Export PDF	<i>Reports</i>	Generate and download PDF report
Create Counterparty	<i>Counterparties</i>	Add new API client/counterparty with credentials
Rotate API Key	<i>Counterparties</i>	Generate new API key for selected counterparty

Enable / Disable	<i>Counterparties</i>	Activate or deactivate a counterparty's API access
Initiate Transfer	<i>Outbound Transfers</i>	Create new outbound USDT transfer request
Broadcast	<i>Outbound Transfers</i>	Manually broadcast an approved transfer to the blockchain
Cancel Transfer	<i>Outbound Transfers</i>	Cancel a pending or approved transfer before broadcast
Create Job	<i>M1 Tokenization</i>	Create new EUR→USDT tokenization job
Run FX	<i>M1 Tokenization</i>	Fetch current EUR/USD exchange rate for a job
Mint Request	<i>M1 Reserve</i>	Submit mint request for new M1F or SIG tokens
Redeem Request	<i>M1 Reserve</i>	Submit redeem request to burn tokens and release reserve
New Snapshot	<i>M1 Reserve</i>	Create new reserve snapshot with proof hash
Top Up Card	<i>Top-Up Engine</i>	Add USDT balance to selected prepaid card
Add Card	<i>Top-Up Engine</i>	Register new prepaid card under a wallet
Add Wallet	<i>Top-Up Engine</i>	Create new top-up wallet
Send SWIFT	<i>SWIFT Terminal</i>	Compose and send SWIFT payment instruction
Create Order	<i>Payments / Stripe</i>	Create new payment checkout session
Refund	<i>Orders / Stripe</i>	Initiate refund for completed order
View on Etherscan	<i>Transfers, Alchemy</i>	Open transaction on Etherscan block explorer
Copy Address	<i>Payments</i>	Copy wallet address to clipboard
Generate QR	<i>Direct Crypto</i>	Generate QR code for wallet address
Ban IP	<i>Security</i>	Manually ban an IP address
Unban IP	<i>Security</i>	Remove manual ban from an IP address

13.3 Private Report Feature

The Private Report feature generates an encrypted, password-protected PDF containing full transaction details for a specific settlement payload or payment order. The report includes: counterparty identity, transaction hash, blockchain proof, review notes, audit trail, and compliance flags. Access is restricted to admin users and the PDF is watermarked CONFIDENTIAL. The button 'Print Private Report ■' appears on the payload detail page.



TOP-UP ENGINE

Prepaid Card Wallets · Card Management · Transactions

14.1 Architecture

The Top-Up Engine manages a hierarchy of TopUpWallet → TopUpCard → TopUpTransaction. Internal wallets hold USDT balances on-chain. Cards are prepaid payment instruments linked to wallets. Transactions record every top-up request from providers.

14.2 Data Models

Model	Key Fields	Purpose
TopUpWallet	name, currency, balance, network, blockchain_address	Holds USDT for card funding
TopUpCard	card_number, holder_name, provider_name, status	Prepaid card linked to wallet
TopUpTransaction	reference, amount, currency, status, provider_ref	Records each top-up event

14.3 Transaction Statuses

- › **PENDING**: Top-up request received, awaiting processing
- › **SUCCESS**: Card successfully topped up
- › **FAILED**: Top-up failed — see failure_reason
- › **REJECTED**: Top-up rejected by compliance review

15

API REFERENCE

All REST Endpoints · Schemas · Responses

15.1 Core Endpoints

Method	Path	Auth	Description
POST	/api/v1/oauth/token	Client credentials	Get OAuth 2.0 access token
POST	/api/v1/payloads/ingest	API Key + HMAC	Submit inbound settlement payload
GET	/api/v1/payloads/schema	API Key	Get payload JSON schema
GET	/api/v1/payloads/{id}	Admin	Get specific payload record
GET	/api/v1/payloads	Admin	List all payloads (paginated)
POST	/api/v1/transfers	Admin	Create outbound USDT transfer
GET	/api/v1/transfers	Admin	List outbound transfers
GET	/api/v1/transfers/{id}	Admin	Get specific transfer
POST	/api/v1/transfers/{id}/approve	Admin	Approve transfer for broadcast
DELETE	/api/v1/transfers/{id}	Admin	Cancel pending transfer
GET	/api/v1/transactions/{tx_id}	Admin	Look up blockchain transaction
POST	/api/v1/transfer-request	API Key	External counterparty transfer request
GET	/api/v1/payments/ledger/status/{tx}	Admin	Check ledger payment status
POST	/api/v1/webhooks/stripe	Stripe-Sig	Stripe webhook receiver
POST	/api/v1/webhooks/alchemy	HMAC	Alchemy settlement webhook
POST	/api/v1/webhooks/circle	API	Circle USDC webhook
POST	/api/v1/webhooks/coinbase	HMAC	Coinbase webhook
POST	/api/v1/webhooks/moonpay	HMAC	MoonPay webhook
POST	/webhooks/alchemy	HMAC	Alchemy event webhook
GET	/health	Token/IP	System health check
GET	/api/v1/admin/clients	Admin	List all API clients
POST	/api/v1/admin/clients	Admin	Create new API client
GET	/api/v1/admin/logs	Admin	Get audit logs
GET	/api/v1/admin/m1/reserve	Admin	Get M1 reserve status
POST	/api/v1/admin/m1/mint	Admin	Submit M1 mint request

POST	/api/v1/admin/m1/redeem	Admin	Submit M1 redeem request
GET	/api/v1/admin/topup/wallets	Admin	List top-up wallets
POST	/api/v1/admin/topup/wallets	Admin	Create top-up wallet
POST	/api/v1/admin/topup/cards	Admin	Create prepaid card
POST	/api/v1/admin/topup/topup	Admin	Process top-up transaction

16

SWIFT TERMINAL

SWIFT Message Composer · Payment Instructions

16.1 Overview

The SWIFT Terminal (/swift) provides a web interface for composing and sending SWIFT financial message instructions. It is accessible separately from the main dashboard sidebar (indicated by the ■ icon). The terminal allows authorized operators to draft MT103 (single customer credit transfer) and MT202 (financial institution transfer) messages, validate field formats, and dispatch to correspondent banks.

16.2 SWIFT Features

- › MT103 single customer credit transfer composer
- › MT202 financial institution transfer composer
- › BIC/IBAN field validation
- › Message draft saving and retrieval
- › Dispatch confirmation logging
- › Integration with audit trail (all SWIFT actions logged)
- › Admin-only access (ADMIN_API_KEY required)

17

CLIENT PORTAL

External Client-Facing Portal · Payment Pages

17.1 Overview

The Client Portal is a separate web application served from the same FastAPI backend. It provides end-customers and counterparties with: order status lookup, payment initiation via multiple providers, transaction history, and document download. The portal is served at the PUBLIC_APP_URL and uses OAuth 2.0 client authentication.

17.2 Portal Pages

- › **Pay / Checkout:** Provider selection: Coinbase, MoonPay, Circle USDC, Stripe, Direct Crypto
- › **Order Status:** Real-time order status with blockchain confirmation progress
- › **Payment Success:** Redirect landing page after successful payment (/pay/success)
- › **Transaction History:** Client's own order history (authenticated)
- › **Documents:** Download transaction receipts and settlement proofs
- › **FNFCU Submit:** Submit structured settlement payloads via FNFCU V1.0 protocol

18

DEPLOYMENT & INFRASTRUCTURE

Render.com · GitHub CI/CD · Redis · Celery · PostgreSQL

18.1 Hosting Architecture

Component	Platform / Service	Details
Web API	Render.com Web Service	Auto-deploy on git push to main branch
Database	PostgreSQL (Render / Supabase)	Async via asyncpg + SQLAlchemy 2.0
Dev Database	SQLite (aiosqlite)	For local development only
Task Queue	Celery	Background jobs: confirmation monitoring, FX fetching
Message Broker	Redis	Celery broker + result backend
Blockchain RPC	Alchemy	ETH mainnet, Base mainnet RPC endpoints
Tron RPC	TronGrid API	TRC-20 USDT monitoring
Error Tracking	Sentry	Exception monitoring and alerting
Email	SMTP/TLS	Notifications via NOTIFY_FROM_EMAIL
Source Control	GitHub	main branch → auto-deploy to Render
Secrets	Environment Variables	All keys via Render environment config

18.2 Environment Variables — Critical Settings

The following environment variables are required for production operation. All secrets are configured via Render's environment variable panel and never committed to source control.

Variable	Required	Purpose
DATABASE_URL	■	PostgreSQL connection string
REDIS_URL	■	Redis connection for Celery
ADMIN_API_KEY	■	Dashboard access key
ALCHEMY_API_KEY	■	Alchemy API access
ALCHEMY_WEBHOOK_SIGNING_KEY	■	Webhook HMAC verification
ALCHEMY_ETHEREUM_RPC_URL	■	Ethereum RPC for settlement
ETH_PRIVATE_KEY	■	Operator wallet signing key
SETTLEMENT_JWE_PRIVATE_KEY_PEM	Opt	JWE decryption (enterprise tier)
STRIPE_SECRET_KEY	Opt	Stripe payment processing
STRIPE_WEBHOOK_SECRET	Opt	Stripe webhook verification

MOONPAY_API_KEY	Opt	MoonPay on-ramp
CIRCLE_API_KEY	Opt	Circle USDC wallet
COINBASE_API_KEY_ID	Opt	Coinbase Commerce
TRUSTED_PROXY_IPS	■	Trusted reverse proxy IPs
PUBLIC_BASE_URL	■	https://api.alshumookh-pay.com
MASTER_WALLET_ETHEREUM	■	Treasury wallet address (ETH)

18.3 Background Tasks (Celery)

- › Transfer confirmation monitor: polls Alchemy every 60s for pending outbound transfers
- › M1 FX rate fetcher: fetches EUR/USD rate for queued tokenization jobs
- › Webhook retry: retries failed webhook callbacks up to 3 times
- › Audit log cleanup: archives old audit records to cold storage
- › Reserve snapshot: scheduled daily reserve valuation snapshot

SYSTEM SUMMARY

Category	Detail
Platform Name	ALSHUMOOKH GLOBAL BANKING FINANCE & CREDIT
API URL	https://api.alshumookh-pay.com
Framework	FastAPI (Python) · Async Architecture
Networks	Ethereum Mainnet · Base Mainnet · Tron Mainnet
Supported Assets	USDT · USDC · ETH · SIG
SIG Token Contract	0xc2AC880E474c3576CC3afB7c560E402ce24d5B37
M1 Token Contract	0xD999DB972BBDc2C13a9595A1474A04F5e59169a5
SIG Issuance Price	\$0.053266 USD per SIG
Payment Providers	Coinbase · MoonPay · Circle · Stripe · Onramper
Blockchain Provider	Alchemy (ETH + Base RPC + Webhooks)
Settlement Protocol	FNFCU V1.0 — 4 security tiers
Auth Methods	API Key · HMAC · OAuth 2.0 · JWS · JWE · mTLS
Security	WAF · Rate Limiting · IP Banning · Challenge Mode
Dashboard Pages	21 pages with 35+ action buttons
Confirmations Required	12 Ethereum blocks (~3 minutes)
Hosting	Render.com · GitHub CI/CD · PostgreSQL · Redis
Audit	100% of protected API calls logged to AuditLog
Document	Comprehensive System Guide v2.0 · June 2026